Pavel Makhov

Staff Software Developer // GenAI Platform, Java, Python

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Career



Staff Software Developer
Lightspeed Commerce

Montreal, Canada
08.2021 - current / 5y 2m

As part of the Agent Squad, I architect and build an agentic AI platform that helps teams deliver production-grade generative features from design and implementation to rollout and operations.

Selected accomplishments:

- Drove the decomposition of a monolithic application into domain-aligned Spring Boot microservices to improve scalability and team autonomy.
- Led the technical design and delivery of a platform-wide authorization system based on the Attribute-Based Access Control (ABAC) model.
- Heavily involved in the architecture and design of new APIs and surrounding tooling, advocating design-first API development and continuously applying modern backend best practices.
- Built an event-driven framework for AI agents using Java, Spring AI, Kafka, Kubernetes, and DynamoDB, integrating Azure AI and Vertex AI APIs for production workflows.
- Passionate about developer tools; built a native macOS menubar application to help engineers manage internal Kubernetes clusters and reproducible development environments, and created multiple OpenCode plugins and internal tools to automate repetitive tasks and improve developer experience.
- Developed a Datadog-based observability framework for LLM features to track latency, cost-per-request, and model-quality signals, enabling faster debugging and safer rollouts.
- Provided expert-level 3rd line support and actively participated in incident on-call rotations, postmortems, and end-to-end resolution of high-impact production issues, including post-incident improvements.
- Partnered with product and engineering stakeholders in discovery and design sessions to define AI feature requirements, delivery plans, and integration constraints.



Senior Software Developer
Solution Media 360

Montreal, Canada
01.2021 - 08.2021 / 7m

Designed and developed Kotlin-based backend services for a leading Canadian provider of innovative technology solutions for the automotive dealership market. Collaborated closely with cross-functional teams to deliver robust and scalable solutions.

Selected accomplishments:

- Implemented an edge-level authorization mechanism for a set of microservices.
- Provided level 3 application support.
- Onboarded and mentored new team members.



Java Developer
Behavox

Montreal, Canada
01.2020 - 01.2021 / 1y

Worked in a product team building compliance management software solutions, developing data-intensive applications using Java, Kotlin, HBase, and MySQL.

Selected accomplishments:

- Led technical design, implementation, and delivery of complex data-intensive applications.



Backend Developer
TouchTunes / OctaveGroup

Montreal, Canada
06.2018 - 12.2019 / 1y 6m

Worked as a Java Backend Developer in the product team for the largest in-venue interactive music and entertainment platform, deployed in over 75,000 bars and restaurants across North America and Europe. Built services in a microservice architecture, with Spring Boot-based applications deployed to AWS in Docker containers managed by Mesos/Marathon. Technology stack: Java 11, Spring Boot, Reactor, Cucumber, MongoDB.

Selected accomplishments:

- Implemented backend integration for Apple Music For Business.
- Developed the Access Control System, a set of microservices implementing the ABAC authorization model.
- Developed multiple payment integrations with Worldpay/Vantiv, PayPal, Apple Pay, and Google Pay.

Full Stack Java Developer
Intelerad

Montreal, Canada
01.2018 - 05.2018 / 5m

Contributed to development of PACS (Picture Archiving and Communication System). My primary focus was Tech Portal, a customization tool that allows technologists to create multiple worklists based on criteria such as exam status, date range, modality, organizations, locations, and rooms. Tech stack: GWT, CSS/SCSS, AngularJS.

Java Developer / Consultant
Savoir-Faire Linux

Montreal, Canada
08.2016 - 12.2017 / 1y 5m

Successfully delivered multiple consulting projects, collaborating with diverse clients to build high-quality solutions. Provided Java expertise across different stages of project lifecycles.

Selected accomplishments:

- Developed a high-performance, highly available Music REST API.
- Developed a real-time data processing application based on Apache Mahout and Apache Kafka.
- Developed a WebRTC Liferay portlet and Content Classifier portlet (based on DL4J library).
- Contributed to building a data pipeline for an ML project.

Java Developer
CERN

Geneva, Switzerland
10.2012 - 05.2016 / 3y 9m

Contributed to development of the business-critical eFiles document management system based on Alfresco, an open-source market leader in records management. As an Alfresco developer, I extended and customized the platform and also worked on EDH (Electronic Document Handling), a system with complex workflows covering key business processes at CERN. In this role, I developed EDH documents and improved existing functionality.

Selected accomplishments:

- Integrated Alfresco with AppDynamics, an application performance management system.
- Developed a web application for 2nd/3rd-line support of eFiles. Technology stack included Spring Boot, AngularJS, and Spring Data REST.

Technical skills

Key competencies:

- Java / Kotlin / Python / Lua / Swift
- Spring / Reactor / Hibernate / JUnit / Mockito / Spock
- Oracle, MySQL / Mongo, Redis, DynamoDB, Couchbase
- Datadog / Dynatrace / Sumologic
- TDD / BDD (Cucumber, Spock)
- Google Cloud (Vertex AI) / Azure AI / Kubernetes / Docker
- REST / Design-first APIs / SDK integrations

Talks

API Development: Shifting from Code-First to Design-First

- ConFoo Conference (2026)
- Montreal JUG

Open Source Projects

Awesome WM widgets
Collection of widgets for Awesome WM.
Lua ★ 2263 ⑂ 295

pullBar
macOS application to show GitHub PRs.
Swift ★ 130 ⑂ 13

Jekyll Clean Dark
Theme for Jekyll.
HTML ★ 226 ⑂ 128

Jekyll CV Crafter
Jekyll-powered CV generator.
HTML ★ 108 ⑂ 47

Education

National Research University of Electronic Technology

Master of Computer Science (2007 – 2012)
https://eng.miet.ru/, Moscow, Russia